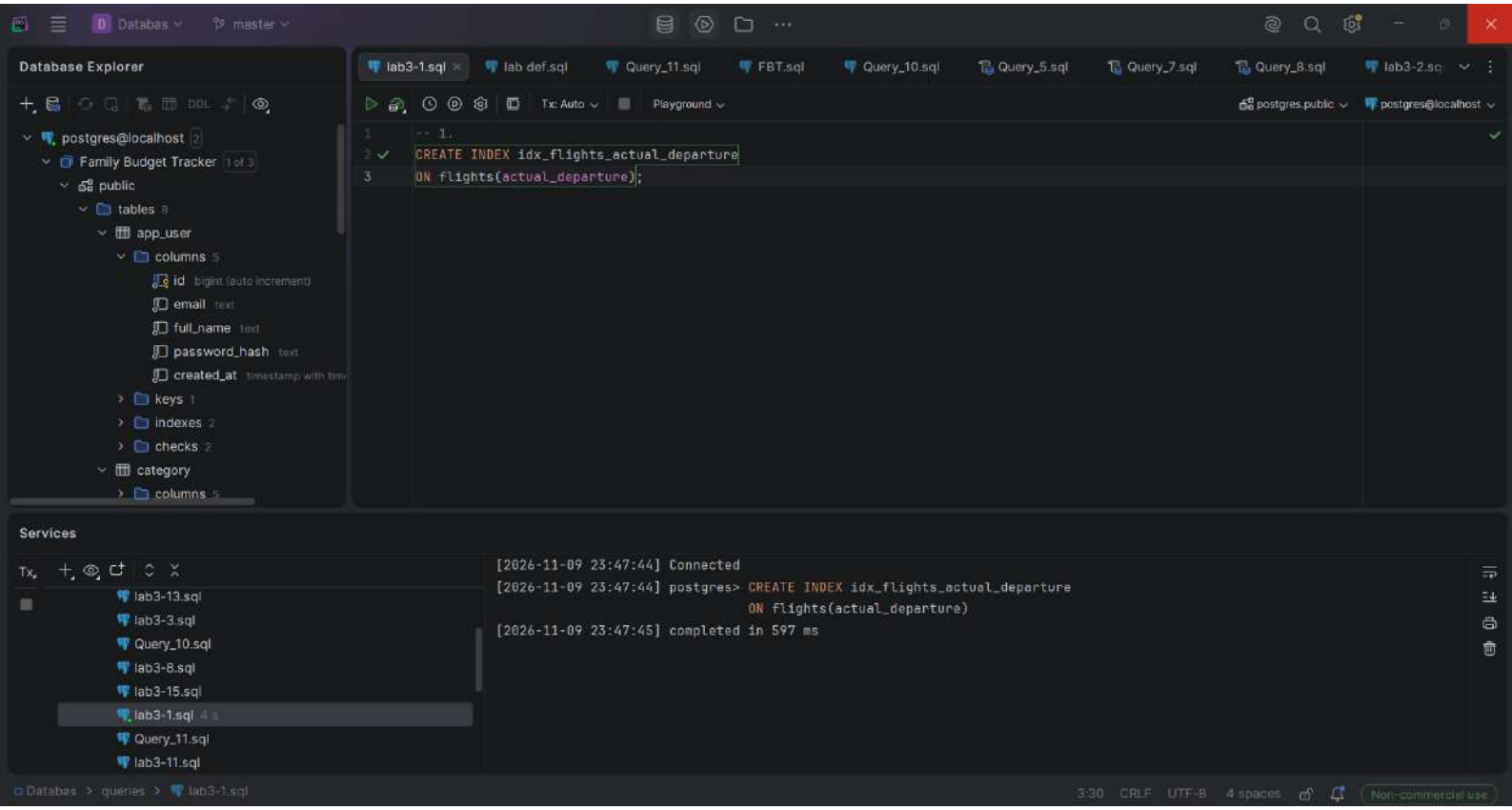
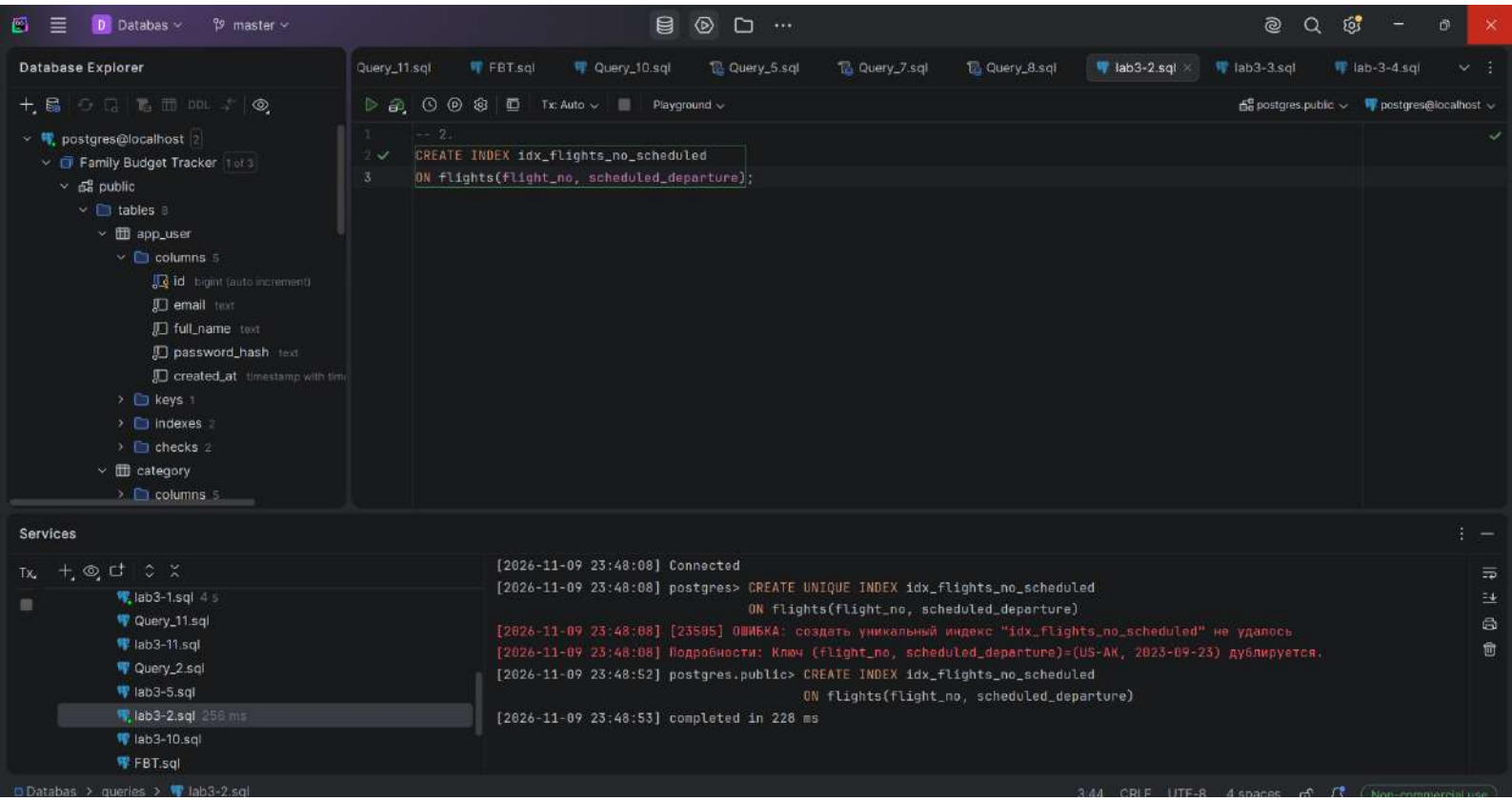
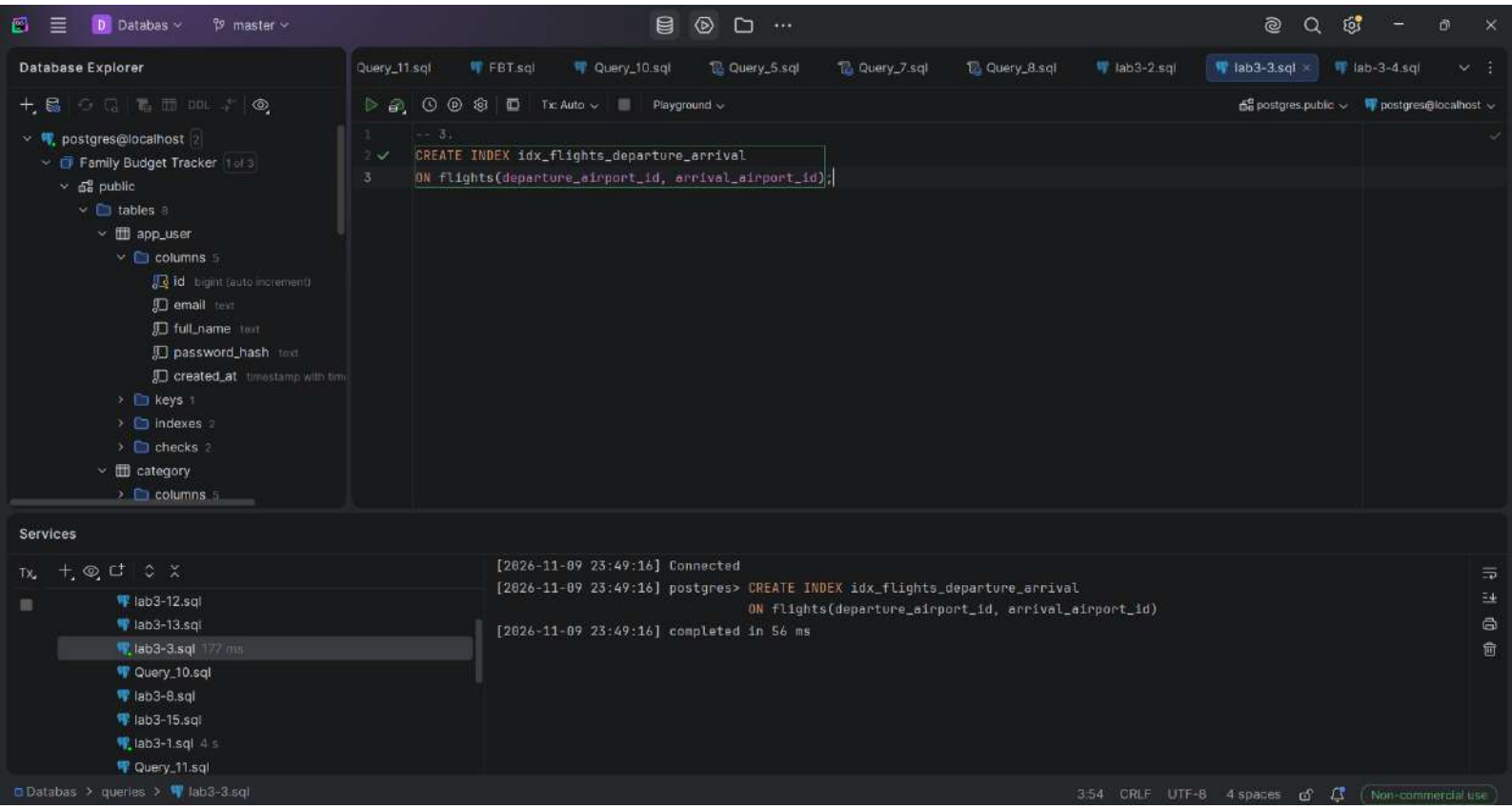
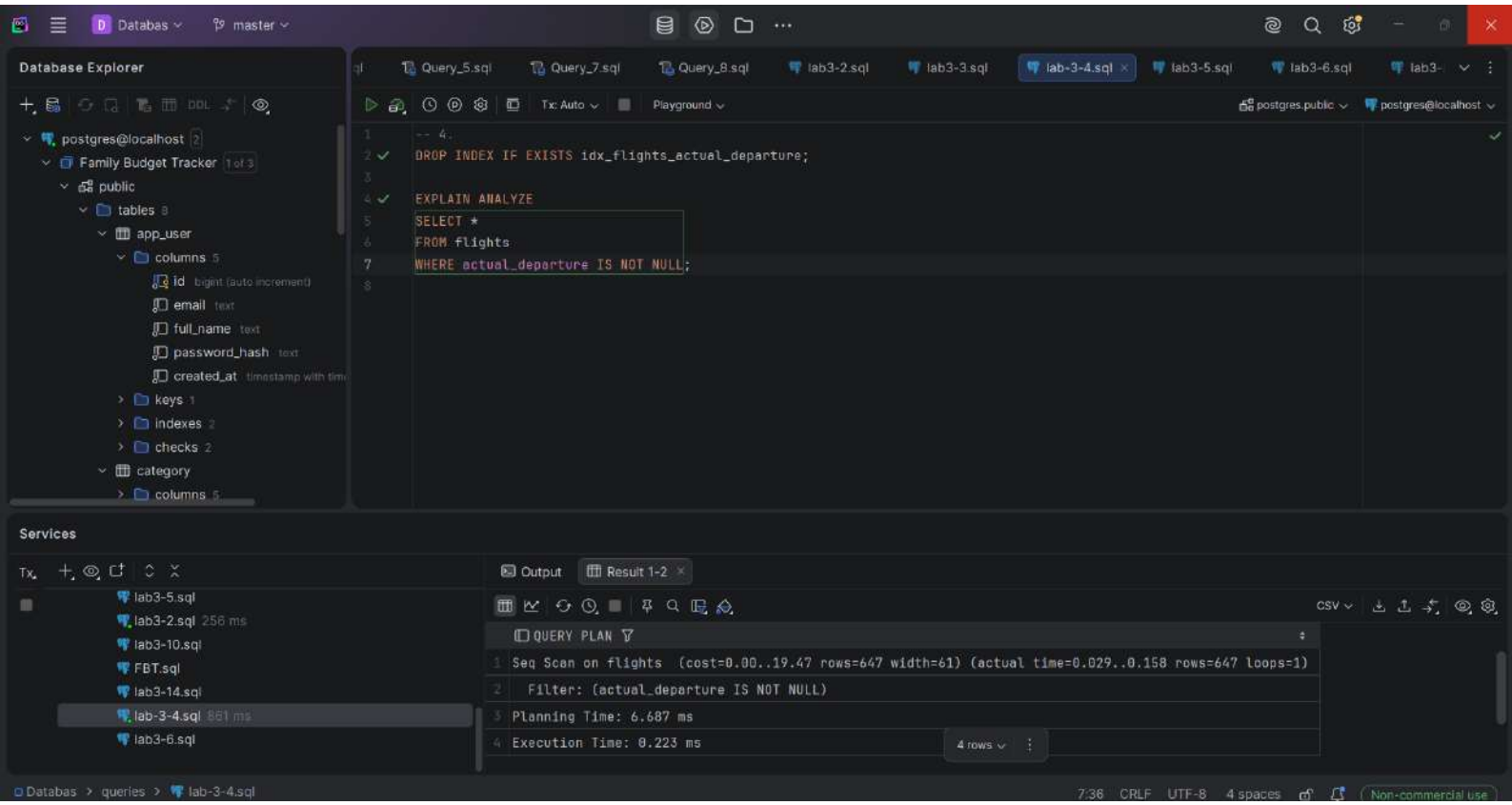
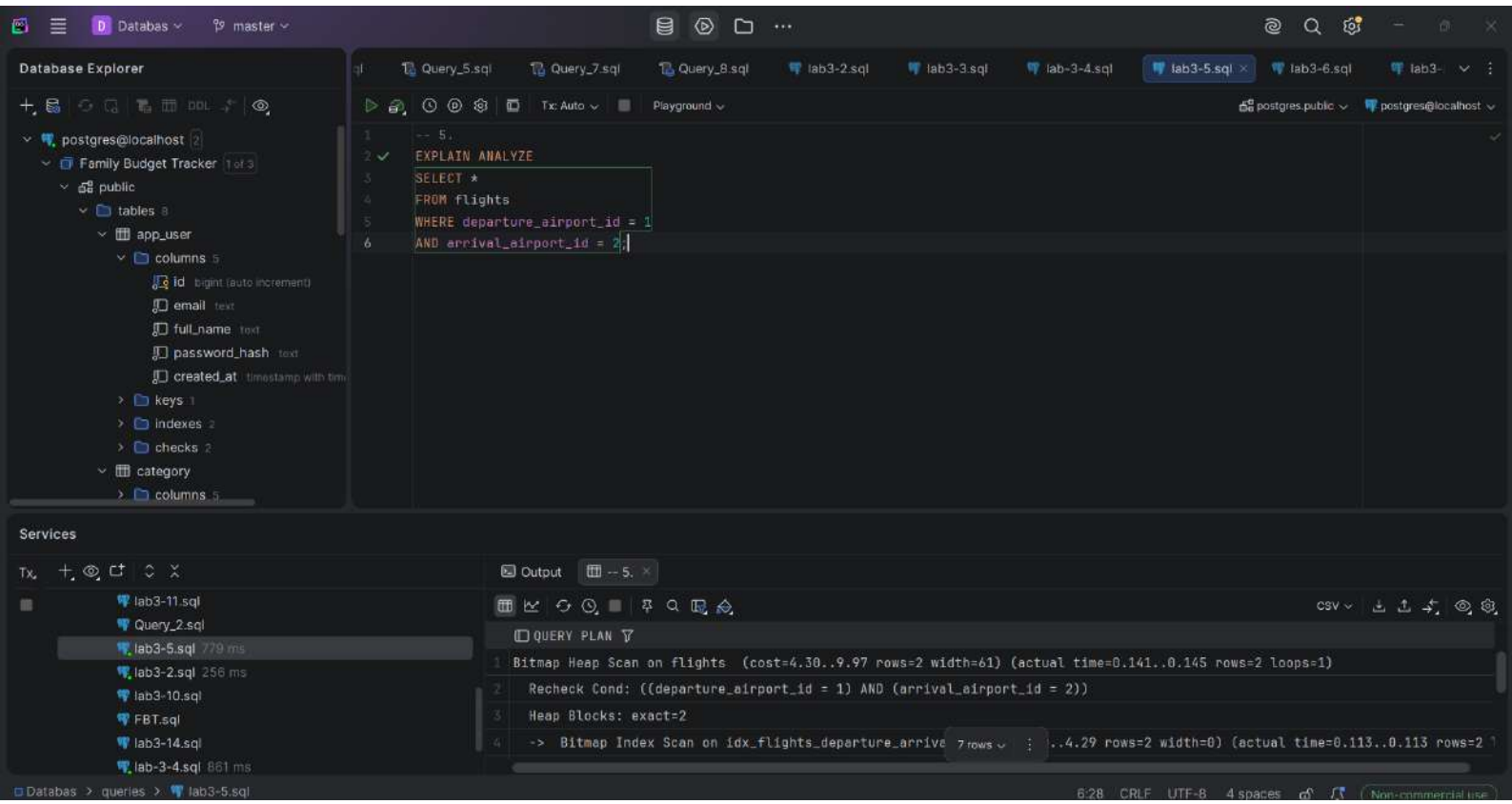
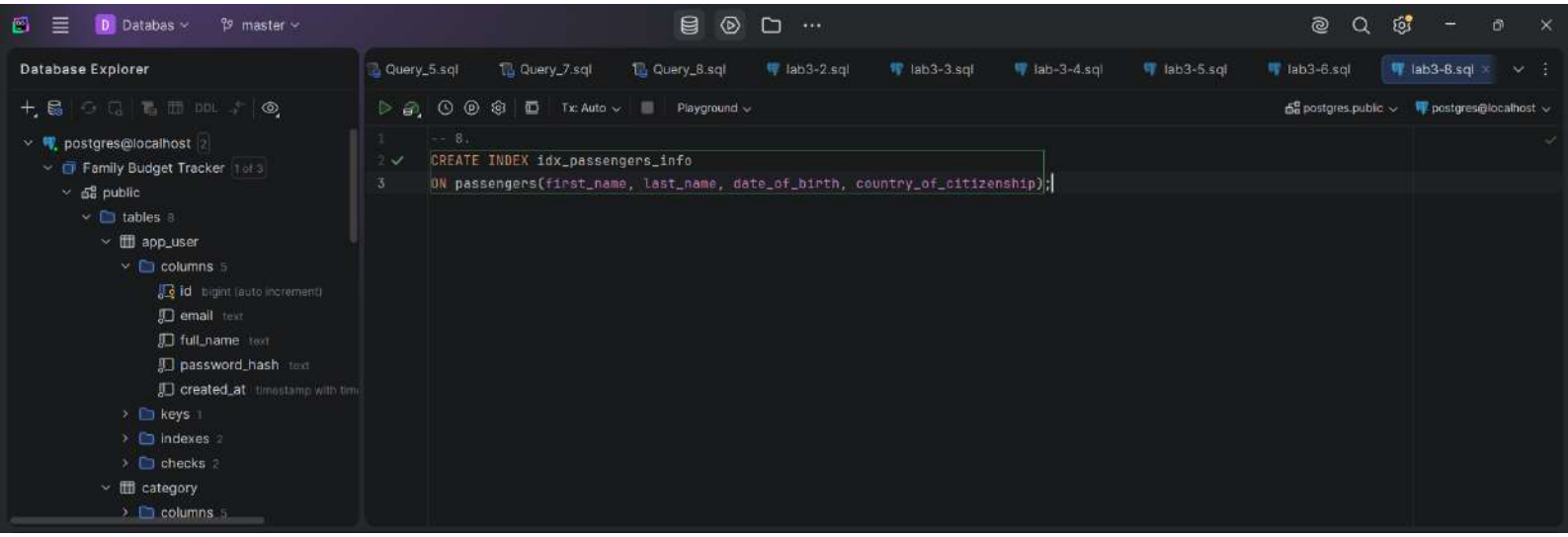

Database Explorer
lab3-1.sql
lab def.sql
Query_11.sql
FBT.sql
Query_10.sql
Query_5.sql
Query_7.sql
Query_8.sql
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id bigint (auto increment)
email text
full_name text
password_hash text
created_at
keys
indexes
checks
category
columns
-- 1.
CREATE INDEX idx_flights_actual_departure
ON flights(actual_departure);
Tx: Auto
Playground
postgres.public
Services
lab3-13.sql
lab3-3.sql
Query_10.sql
lab3-8.sql
lab3-15.sql
lab3-1.sql
Query_11.sql
lab3-11.sql
[2026-11-09 23:47:44] Connected
[2026-11-09 23:47:44] postgres> CREATE INDEX idx_flights_actual_departure
ON flights(actual_departure)
[2026-11-09 23:47:45] completed in 597 ms
Databas > queries > lab3-1.sql
3:30 CRLF UTF-8 4 spaces
Non-commercial use

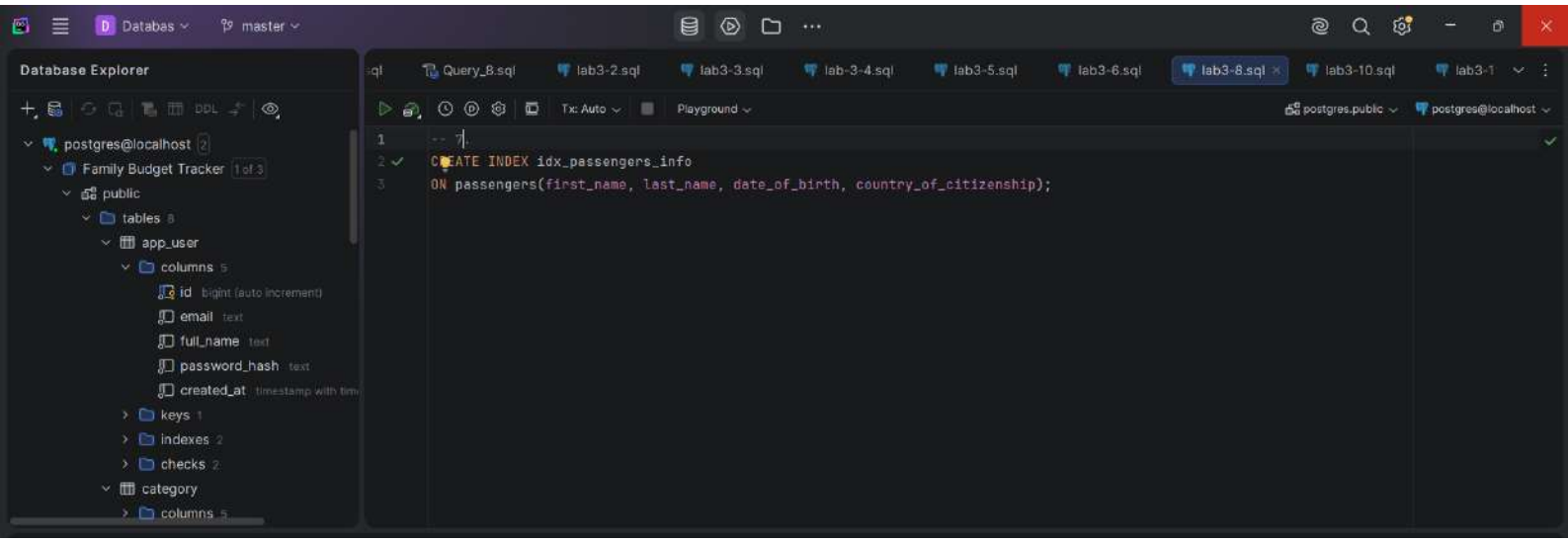

Databas
master
Database Explorer
Query_11.sql
FBT.sql
Query_10.sql
Query_5.sql
Query_7.sql
Query_8.sql
lab3-2.sql
lab3-3.sql
lab-3-4.sql
Tx: Auto
Playground
postgres.public
postgres@localhost
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id
email
full_name
password_hash
created_at
keys
indexes
checks
category
columns
-- 2.
CREATE INDEX idx_flights_no_scheduled
ON flights(flight_no, scheduled_departure);
Services
lab3-1.sql
Query_11.sql
lab3-11.sql
Query_2.sql
lab3-5.sql
lab3-2.sql
lab3-10.sql
FBT.sql
[2026-11-09 23:48:08] Connected
[2026-11-09 23:48:08] postgres> CREATE UNIQUE INDEX idx_flights_no_scheduled
ON flights(flight_no, scheduled_departure)
[2026-11-09 23:48:08] [23505] ОШИБКА: создать уникальный индекс "idx_flights_no_scheduled" не удалось
[2026-11-09 23:48:08] Подробности: Ключ (flight_no, scheduled_departure)=(US-AK, 2023-09-23) дублируется.
[2026-11-09 23:48:52] postgres.public> CREATE INDEX idx_flights_no_scheduled
ON flights(flight_no, scheduled_departure)
[2026-11-09 23:48:53] completed in 228 ms
Databas > queries > lab3-2.sql
3:44
CRLF
UTF-8
4 spaces

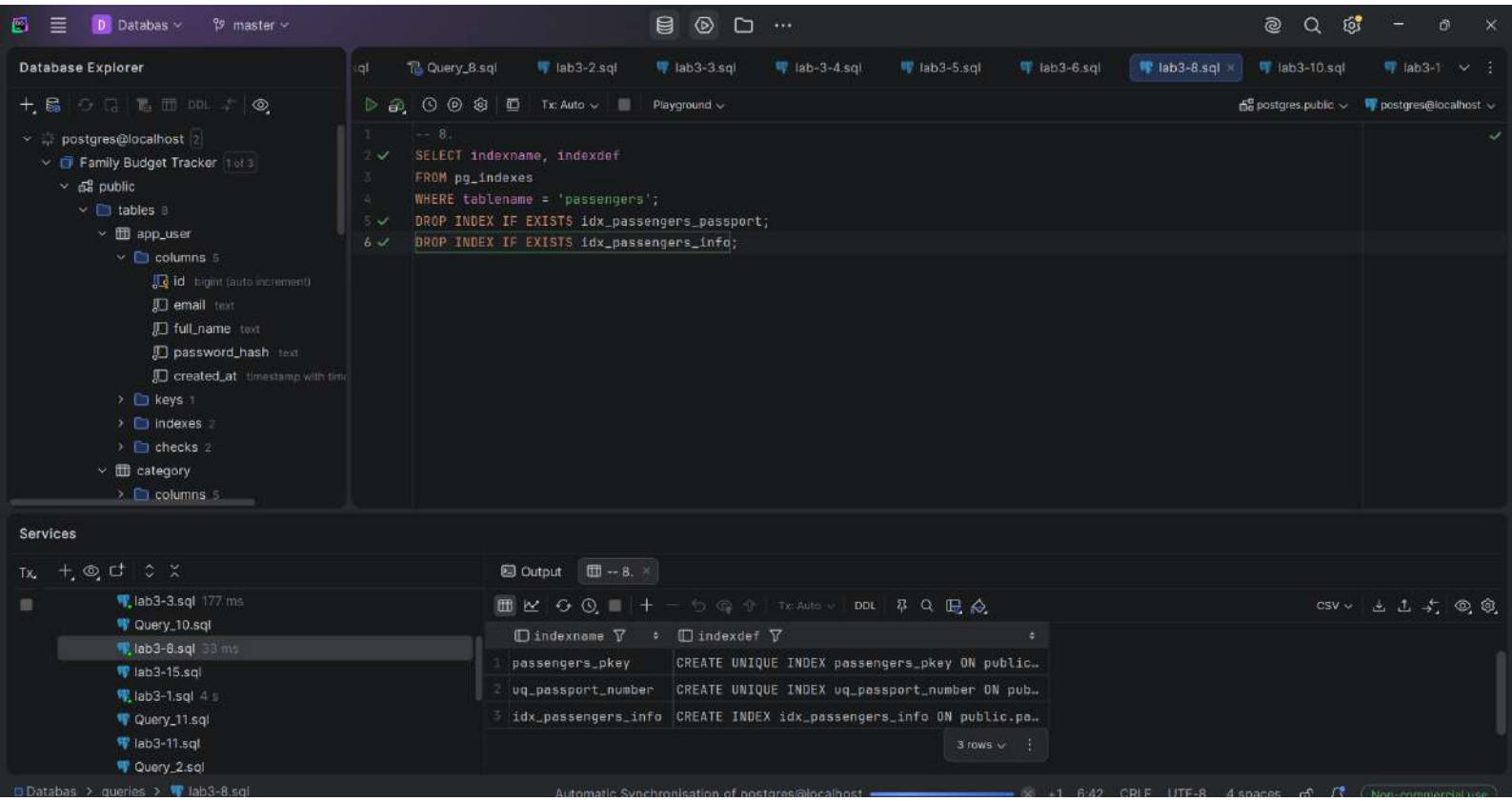

Databas
master
Database Explorer
Query_11.sql
FBT.sql
Query_10.sql
Query_5.sql
Query_7.sql
Query_8.sql
lab3-2.sql
lab3-3.sql
lab-3-4.sql
Tx: Auto
Playground
postgres.public
postgres@localhost
-- 3.
CREATE INDEX idx_flights_departure_arrival
ON flights(departure_airport_id, arrival_airport_id);
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id
email
full_name
password_hash
created_at
keys
indexes
checks
category
columns
Services
lab3-12.sql
lab3-13.sql
lab3-3.sql
Query_10.sql
lab3-8.sql
lab3-15.sql
lab3-1.sql
Query_11.sql
[2026-11-09 23:49:16] Connected
[2026-11-09 23:49:16] postgres> CREATE INDEX idx_flights_departure_arrival
ON flights(departure_airport_id, arrival_airport_id)
[2026-11-09 23:49:16] completed in 56 ms
Databas > queries > lab3-3.sql
3:54
CRLF
UTF-8
4 spaces
Non-commercial use

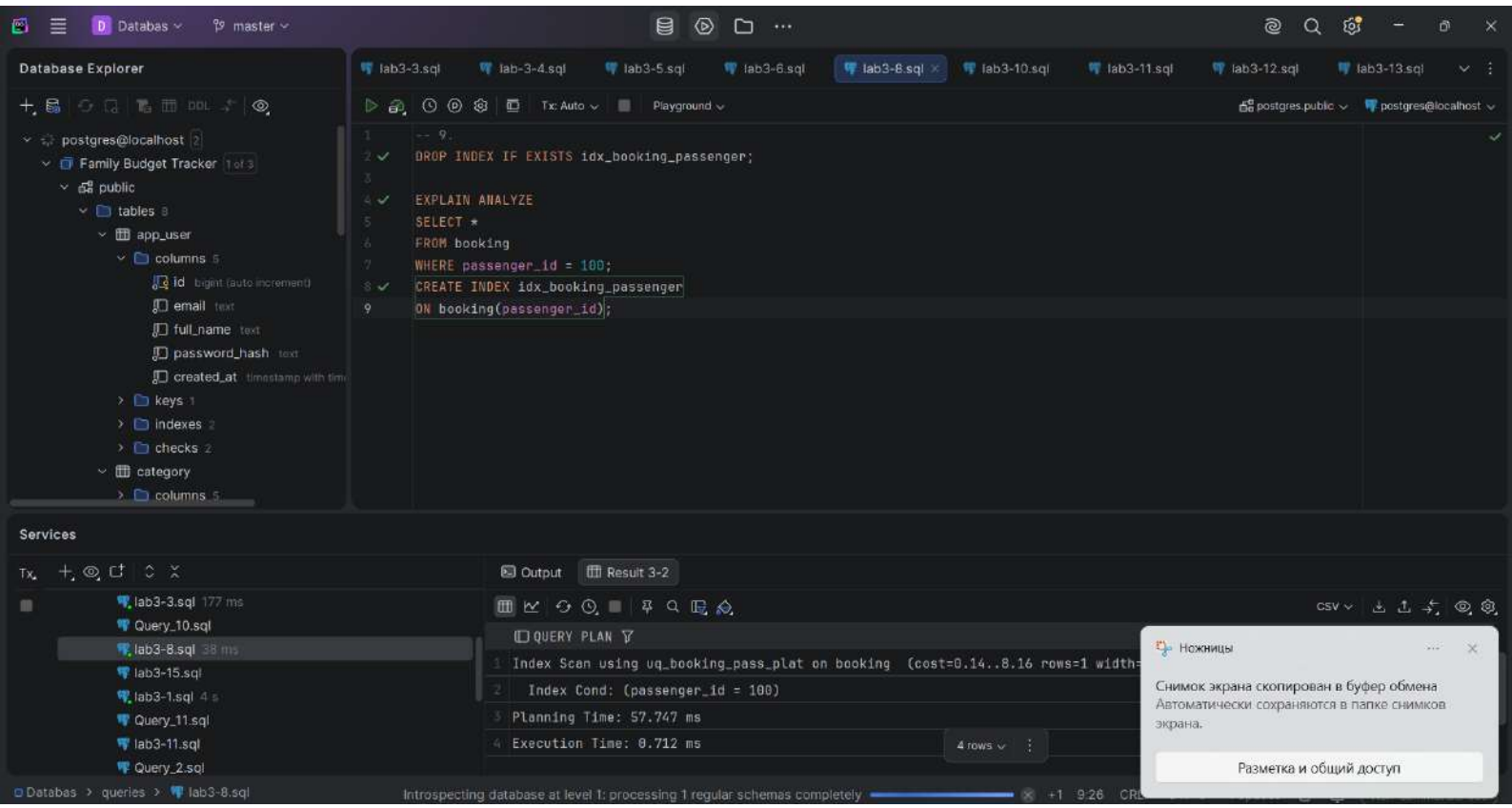

Databas
master
Database Explorer
Query_5.sql
Query_7.sql
Query_8.sql
lab3-2.sql
lab3-3.sql
lab-3-4.sql
lab3-5.sql
lab3-6.sql
Tx: Auto
Playground
postgres.public
postgres@localhost
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id
email
full_name
password_hash
created_at
keys
indexes
checks
category
columns
-- 4.
DROP INDEX IF EXISTS idx_flights_actual_departure;
EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE actual_departure IS NOT NULL;
Services
lab3-5.sql
lab3-2.sql 256 ms
lab3-10.sql
FBT.sql
lab3-14.sql
lab-3-4.sql 861 ms
lab3-6.sql
Output
Result 1-2
CSV
QUERY PLAN
Seq Scan on flights (cost=0.00..19.47 rows=647 width=61) (actual time=0.029..0.158 rows=647 loops=1)
Filter: (actual_departure IS NOT NULL)
Planning Time: 6.687 ms
Execution Time: 0.223 ms
4 rows
Databas > queries > lab-3-4.sql
7:36
CRLF
UTF-8
4 spaces
Non-commercial use

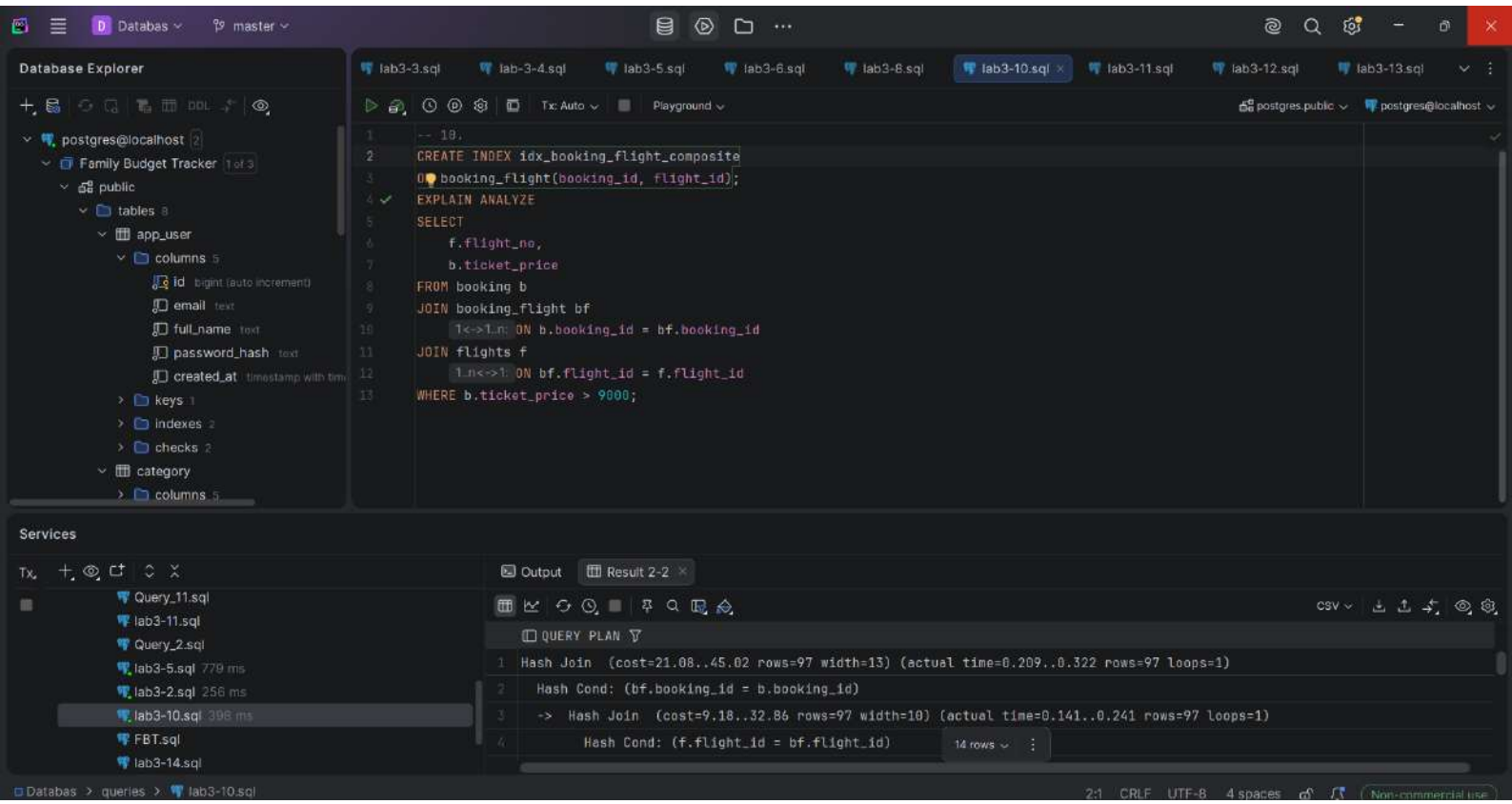

Databas
master
Database Explorer
Query_5.sql
Query_7.sql
Query_8.sql
lab3-2.sql
lab3-3.sql
lab-3-4.sql
lab3-5.sql
lab3-6.sql
Tx: Auto
Playground
postgres.public
postgres@localhost
-- 5.
EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE departure_airport_id = 1
AND arrival_airport_id = 2;
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id
email
full_name
password_hash
created_at
keys
indexes
checks
category
columns
Services
Output
-- 5.
lab3-11.sql
Query_2.sql
lab3-5.sql 779 ms
lab3-2.sql 256 ms
lab3-10.sql
FBT.sql
lab3-14.sql
lab-3-4.sql 861 ms
QUERY PLAN
Bitmap Heap Scan on flights (cost=4.30..9.97 rows=2 width=61) (actual time=0.141..0.145 rows=2 loops=1)
Recheck Cond: ((departure_airport_id = 1) AND (arrival_airport_id = 2))
Heap Blocks: exact=2
-> Bitmap Index Scan on idx_flights_departure_arrive
7 rows
..4.29 rows=2 width=0) (actual time=0.113..0.113 rows=2
Databas > queries > lab3-5.sql
6:28
CRLF
UTF-8
4 spaces
Non-commercial use

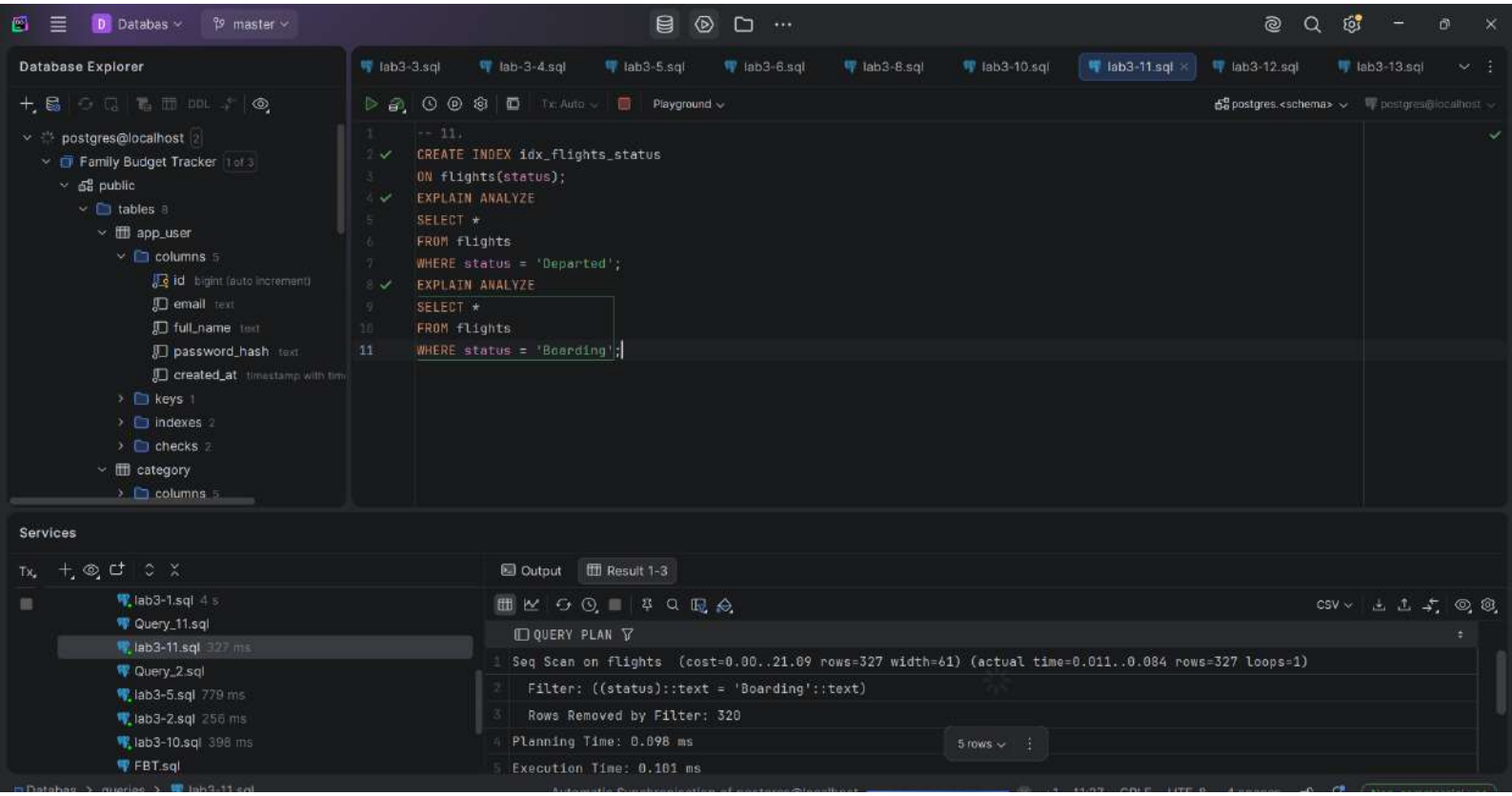

Databas
master
Database Explorer
Query_5.sql
Query_7.sql
Query_8.sql
lab3-2.sql
lab3-3.sql
lab3-4.sql
lab3-5.sql
lab3-6.sql
lab3-8.sql
Tx: Auto
Playground
postgres.public
postgres@localhost
postgres@localhost 2
Family Budget Tracker 1 of 3
public
tables 8
app_user
columns 5
id bigint (auto increment)
email text
full_name text
password_hash text
created_at timestamp with tim
keys 1
indexes 2
checks 2
category
columns 5
-- 8.
CREATE INDEX idx_passengers_info
ON passengers(first_name, last_name, date_of_birth, country_of_citizenship);

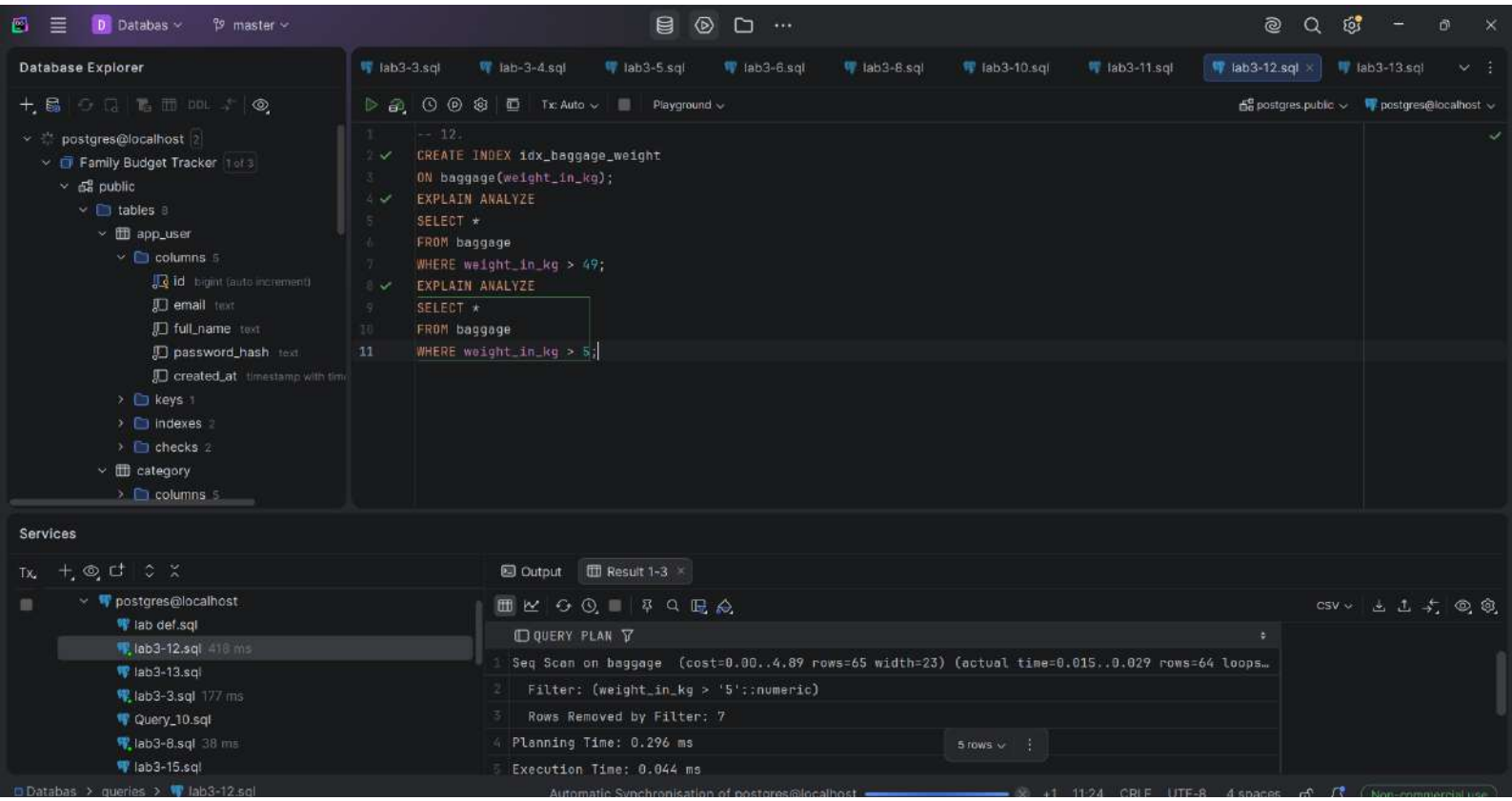

Databas
master
Database Explorer
Query_8.sql
lab3-2.sql
lab3-3.sql
lab-3-4.sql
lab3-5.sql
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-1
Tx: Auto
Playground
postgres.public
postgres@localhost
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id bigint (auto increment)
email text
full_name text
password_hash text
created_at
keys
indexes
checks
category
columns
-- 7.
CREATE INDEX idx_passengers_info
ON passengers(first_name, last_name, date_of_birth, country_of_citizenship);

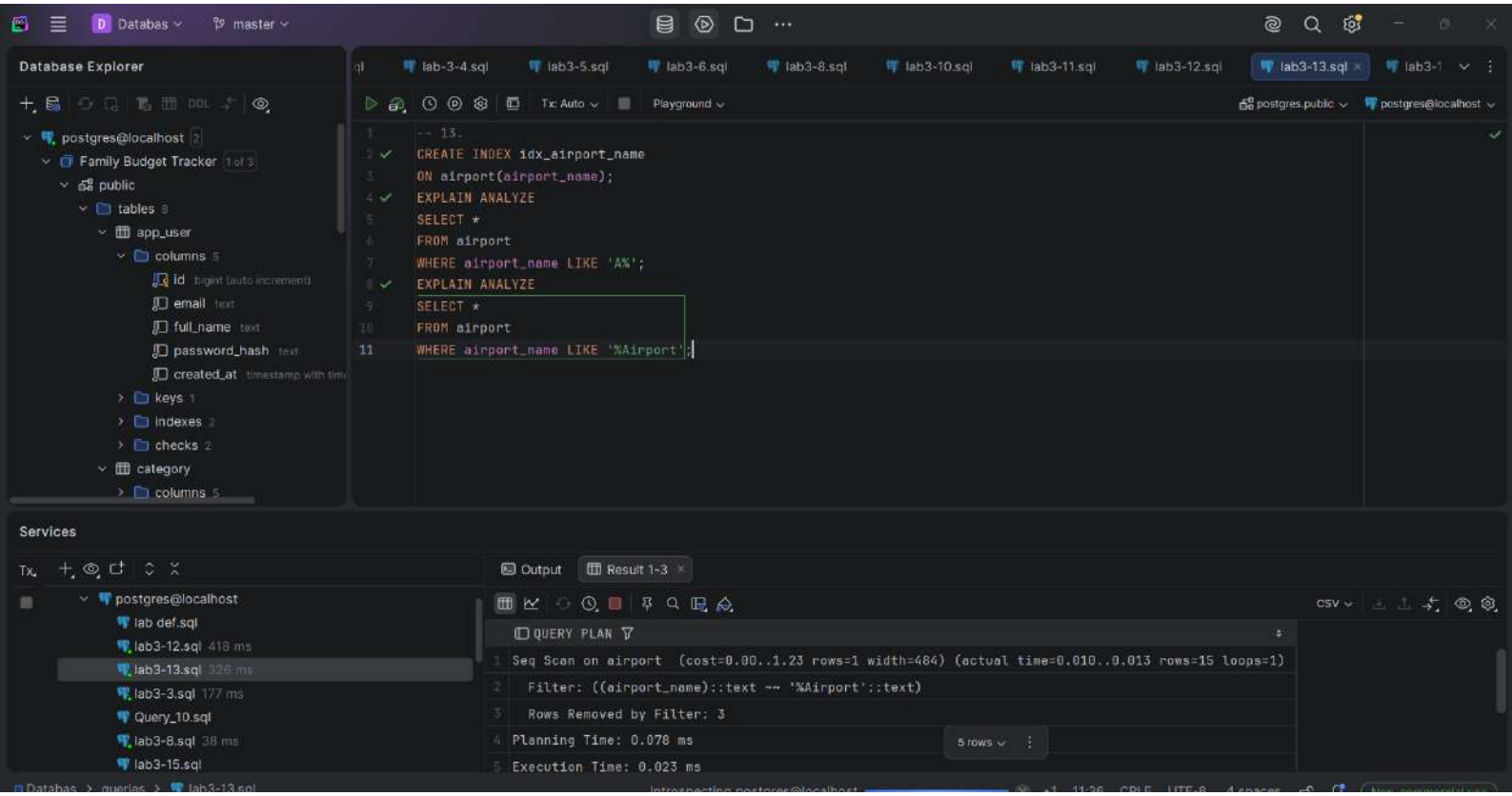

Databas
master
Database Explorer
Query_8.sql
lab3-2.sql
lab3-3.sql
lab-3-4.sql
lab3-5.sql
lab3-6.sql
lab3-8.sql
lab3-10.sql
Tx: Auto
Playground
postgres.public
postgres@localhost
postgres@localhost 2
Family Budget Tracker 1 of 3
public
tables 8
app_user
columns 5
id bigint (auto increment)
email text
full_name text
password_hash text
created_at timestamp with time
keys 1
indexes 2
checks 2
category
columns 5
-- 8.
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'passengers';
DROP INDEX IF EXISTS idx_passengers_passport;
DROP INDEX IF EXISTS idx_passengers_info;
Services
Output
-- 8.
lab3-3.sql 177 ms
Query_10.sql
lab3-8.sql 33 ms
lab3-15.sql
lab3-1.sql 4 s
Query_11.sql
lab3-11.sql
Query_2.sql
indexname
indexdef
1 passengers_pkey CREATE UNIQUE INDEX passengers_pkey ON public…
2 uq_passport_number CREATE UNIQUE INDEX uq_passport_number ON pub…
3 idx_passengers_info CREATE INDEX idx_passengers_info ON public.pa…
3 rows
Databas > queries > lab3-8.sql
Automatic Synchronisation of postgres@localhost
6:42
CRLF
UTF-8
4 spaces
Non-commercial use

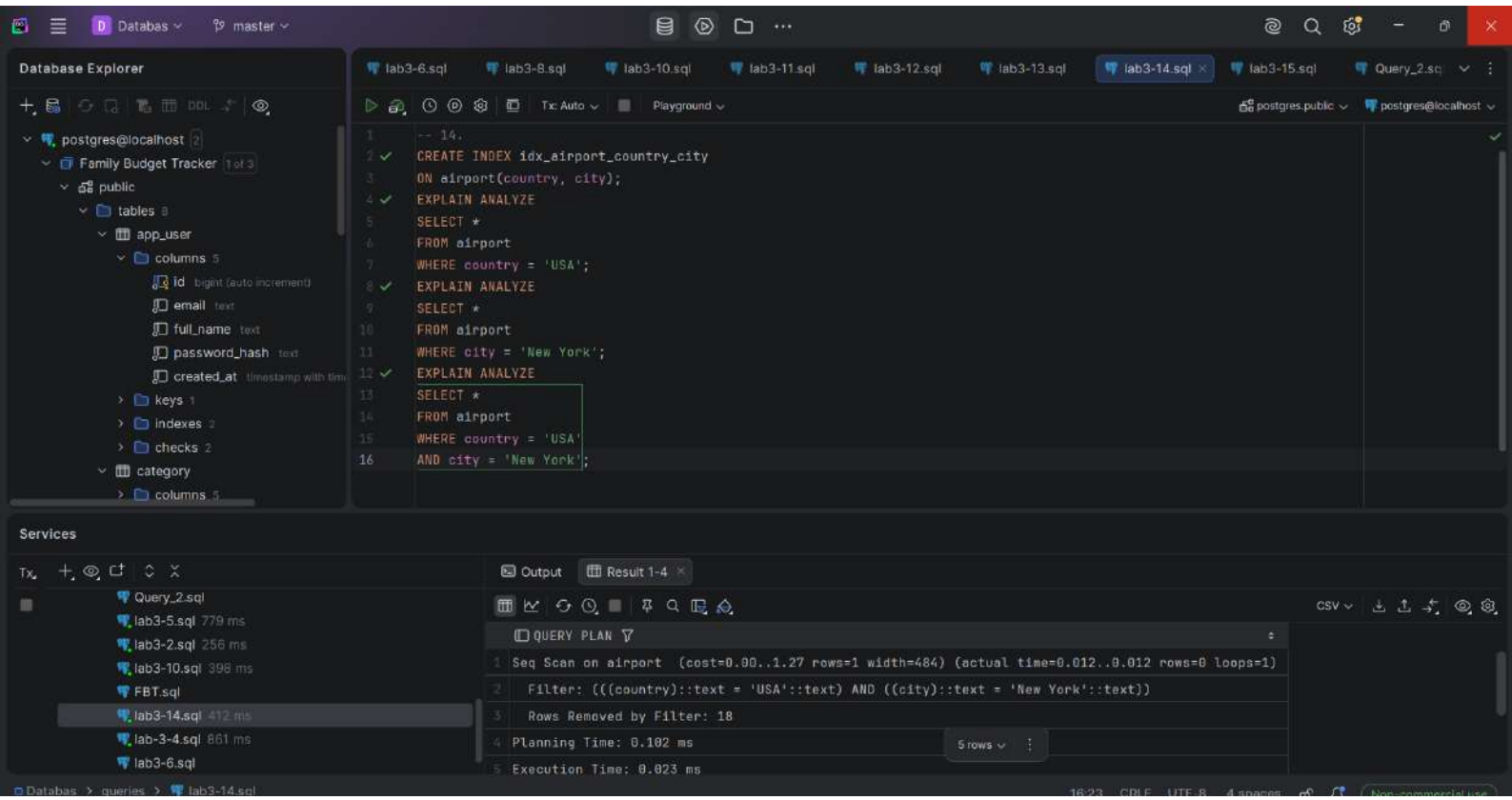

Database Explorer
lab3-3.sql
lab-3-4.sql
lab3-5.sql
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-11.sql
lab3-12.sql
lab3-13.sql
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id
email
full_name
password_hash
created_at
keys
indexes
checks
category
columns
-- 9.
DROP INDEX IF EXISTS idx_booking_passenger;
EXPLAIN ANALYZE
SELECT *
FROM booking
WHERE passenger_id = 100;
CREATE INDEX idx_booking_passenger
ON booking(passenger_id);
Services
Output
Result 3-2
QUERY PLAN
Index Scan using uq_booking_pass_plat on booking (cost=0.14..8.16 rows=1 width
Index Cond: (passenger_id = 100)
Planning Time: 57.747 ms
Execution Time: 0.712 ms
4 rows
lab3-3.sql 177 ms
Query_10.sql
lab3-8.sql 38 ms
lab3-15.sql
lab3-1.sql 4 s
Query_11.sql
lab3-11.sql
Query_2.sql
Ножницы
Снимок экрана скопирован в буфер обмена
Автоматически сохраняются в папке снимков экрана.
Разметка и общий доступ
Databas > queries > lab3-8.sql
Introspecting database at level 1: processing 1 regular schemas completely
9:26

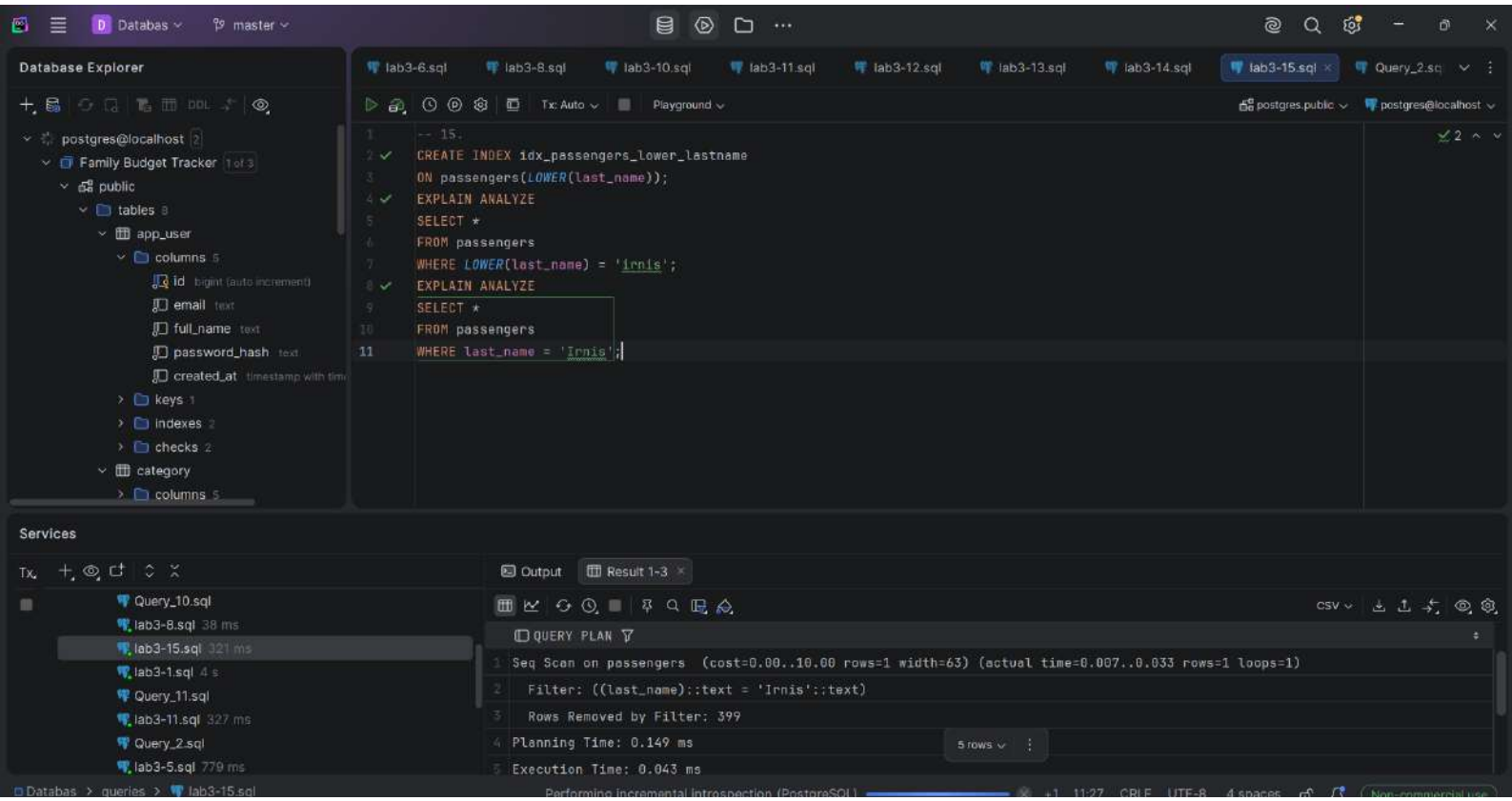

Databas
master
Database Explorer
lab3-3.sql
lab-3-4.sql
lab3-5.sql
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-11.sql
lab3-12.sql
lab3-13.sql
Tx: Auto
Playground
postgres.public
postgres@localhost
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id
email
full_name
password_hash
created_at
keys
indexes
checks
category
columns
-- 10.
CREATE INDEX idx_booking_flight_composite
ON booking_flight(booking_id, flight_id);
EXPLAIN ANALYZE
SELECT
f.flight_no,
b.ticket_price
FROM booking b
JOIN booking_flight bf
1<->1..n: ON b.booking_id = bf.booking_id
JOIN flights f
1..n<->1: ON bf.flight_id = f.flight_id
WHERE b.ticket_price > 9000;
Services
Query_11.sql
lab3-11.sql
Query_2.sql
lab3-5.sql 779 ms
lab3-2.sql 256 ms
lab3-10.sql 396 ms
FBT.sql
lab3-14.sql
Output
Result 2-2
csv
QUERY PLAN
Hash Join (cost=21.08..45.02 rows=97 width=13) (actual time=0.209..0.322 rows=97 loops=1)
Hash Cond: (bf.booking_id = b.booking_id)
-> Hash Join (cost=9.18..32.86 rows=97 width=10) (actual time=0.141..0.241 rows=97 loops=1)
Hash Cond: (f.flight_id = bf.flight_id)
14 rows
Databas > queries > lab3-10.sql
2:1
CRLF
UTF-8
4 spaces
Non-commercial use

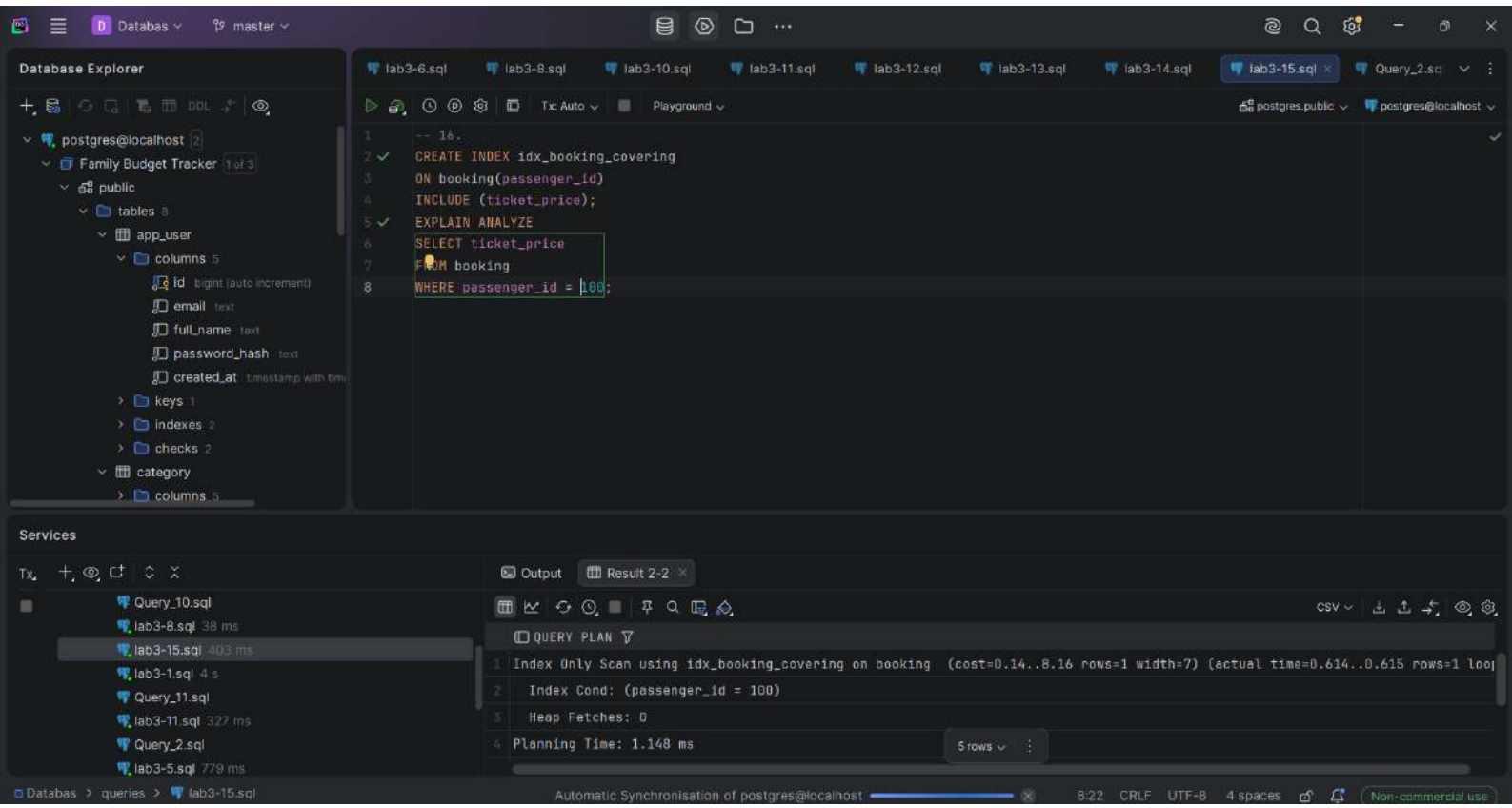

Databas
master
Database Explorer
lab3-3.sql
lab-3-4.sql
lab3-5.sql
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-11.sql
lab3-12.sql
lab3-13.sql
Playground
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id bigint (auto increment)
email text
full_name text
password_hash text
created_at timestamp with tim
keys
indexes
checks
category
columns
-- 11.
CREATE INDEX idx_flights_status
ON flights(status);
EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE status = 'Departed';
EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE status = 'Boarding';
Services
Output
Result 1-3
CSV
lab3-1.sql 4 s
Query_11.sql
lab3-11.sql 327 ms
Query_2.sql
lab3-5.sql 779 ms
lab3-2.sql 256 ms
lab3-10.sql 398 ms
FBT.sql
QUERY PLAN
Seq Scan on flights (cost=0.00..21.09 rows=327 width=61) (actual time=0.011..0.084 rows=327 loops=1)
Filter: ((status)::text = 'Boarding'::text)
Rows Removed by Filter: 320
Planning Time: 0.098 ms
Execution Time: 0.101 ms
5 rows

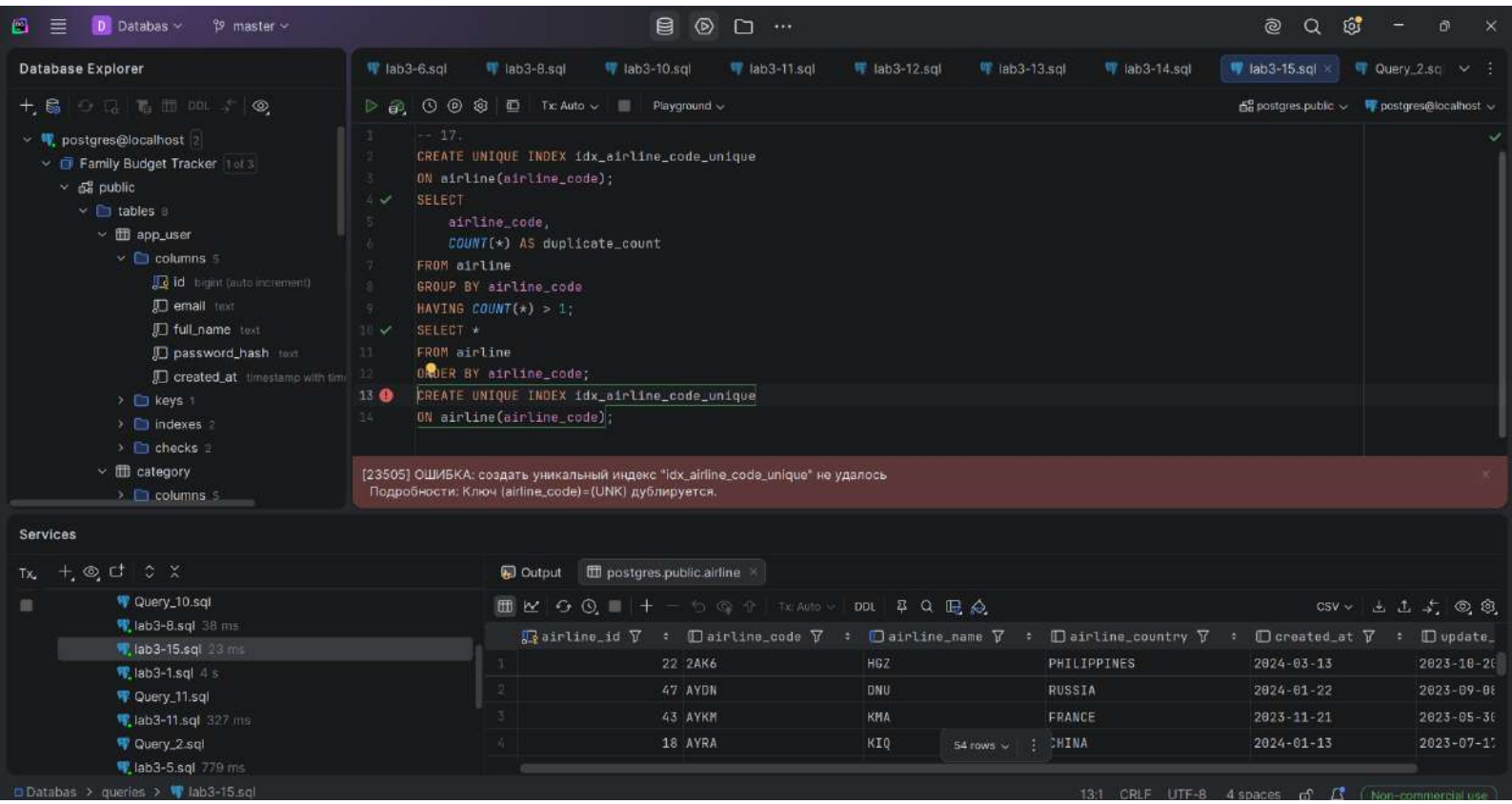

Databas
master
Database Explorer
lab3-3.sql
lab-3-4.sql
lab3-5.sql
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-11.sql
lab3-12.sql
lab3-13.sql
Tx: Auto
Playground
postgres.public
postgres@localhost
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id bigint (auto increment)
email text
full_name text
password_hash text
created_at timestamp with time
keys
indexes
checks
category
columns
-- 12.
CREATE INDEX idx_baggage_weight
ON baggage(weight_in_kg);
EXPLAIN ANALYZE
SELECT *
FROM baggage
WHERE weight_in_kg > 49;
EXPLAIN ANALYZE
SELECT *
FROM baggage
WHERE weight_in_kg > 5;
Services
Output
Result 1-3
CSV
postgres@localhost
lab def.sql
lab3-12.sql 418 ms
lab3-13.sql
lab3-3.sql 177 ms
Query_10.sql
lab3-8.sql 38 ms
lab3-15.sql
QUERY PLAN
Seq Scan on baggage (cost=0.00..4.89 rows=65 width=23) (actual time=0.015..0.029 rows=64 loops…
Filter: (weight_in_kg > '5'::numeric)
Rows Removed by Filter: 7
Planning Time: 0.296 ms
Execution Time: 0.044 ms
5 rows
Databas > queries > lab3-12.sql
Automatic Synchronisation of postgres@localhost
11:24
CRLF
UTF-8
4 spaces
Non-commercial use


Databas
master
Database Explorer
lab-3-4.sql
lab3-5.sql
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-11.sql
lab3-12.sql
lab3-13.sql
Tx: Auto
Playground
postgres.public
postgres@localhost
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id bigint (auto increment)
email text
full_name text
password_hash text
created_at timestamp with tim
keys
indexes
checks
category
columns
-- 13.
CREATE INDEX idx_airport_name
ON airport(airport_name);
EXPLAIN ANALYZE
SELECT *
FROM airport
WHERE airport_name LIKE 'A%';
EXPLAIN ANALYZE
SELECT *
FROM airport
WHERE airport_name LIKE '%Airport';
Services
Output
Result 1-3
csv
postgres@localhost
lab def.sql
lab3-12.sql 418 ms
lab3-13.sql 326 ms
lab3-3.sql 177 ms
Query_10.sql
lab3-8.sql 38 ms
lab3-15.sql
QUERY PLAN
Seq Scan on airport (cost=0.00..1.23 rows=1 width=484) (actual time=0.010..0.013 rows=15 loops=1)
Filter: ((airport_name)::text ~~ '%Airport'::text)
Rows Removed by Filter: 3
Planning Time: 0.078 ms
Execution Time: 0.023 ms
5 rows
Databas > queries > lab3-13.sql


Database Explorer
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-11.sql
lab3-12.sql
lab3-13.sql
lab3-14.sql
lab3-15.sql
Query_2.sq
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id bigint (auto increment)
email text
full_name text
password_hash text
created_at timestamp with time
keys
indexes
checks
category
columns
-- 14.
CREATE INDEX idx_airport_country_city
ON airport(country, city);
EXPLAIN ANALYZE
SELECT *
FROM airport
WHERE country = 'USA';
EXPLAIN ANALYZE
SELECT *
FROM airport
WHERE city = 'New York';
EXPLAIN ANALYZE
SELECT *
FROM airport
WHERE country = 'USA'
AND city = 'New York';
Services
Output
Result 1-4
Query_2.sql
lab3-5.sql 779 ms
lab3-2.sql 256 ms
lab3-10.sql 398 ms
FBT.sql
lab3-14.sql 412 ms
lab-3-4.sql 861 ms
lab3-6.sql
QUERY PLAN
Seq Scan on airport (cost=0.00..1.27 rows=1 width=484) (actual time=0.012..0.012 rows=0 loops=1)
Filter: (((country)::text = 'USA'::text) AND ((city)::text = 'New York'::text))
Rows Removed by Filter: 18
Planning Time: 0.102 ms
Execution Time: 0.023 ms
5 rows
CSV


Database Explorer
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-11.sql
lab3-12.sql
lab3-13.sql
lab3-14.sql
lab3-15.sql
Query_2.sql
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id
email
full_name
password_hash
created_at
keys
indexes
checks
category
columns
-- 15.
CREATE INDEX idx_passengers_lower_lastname
ON passengers(LOWER(last_name));
EXPLAIN ANALYZE
SELECT *
FROM passengers
WHERE LOWER(last_name) = 'irnis';
EXPLAIN ANALYZE
SELECT *
FROM passengers
WHERE last_name = 'Irnis';
Services
Query_10.sql
lab3-8.sql 38 ms
lab3-15.sql 321 ms
lab3-1.sql 4 s
Query_11.sql
lab3-11.sql 327 ms
Query_2.sql
lab3-5.sql 779 ms
Output
Result 1-3
QUERY PLAN
Seq Scan on passengers (cost=0.00..10.00 rows=1 width=63) (actual time=0.007..0.033 rows=1 loops=1)
Filter: ((last_name)::text = 'Irnis'::text)
Rows Removed by Filter: 399
Planning Time: 0.149 ms
Execution Time: 0.043 ms
5 rows


Database Explorer
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-11.sql
lab3-12.sql
lab3-13.sql
lab3-14.sql
lab3-15.sql
Query_2.sq
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id bigint (auto increment)
email text
full_name text
password_hash text
created_at timestamp with tim
keys
indexes
checks
category
columns
-- 16.
CREATE INDEX idx_booking_covering
ON booking(passenger_id)
INCLUDE (ticket_price);
EXPLAIN ANALYZE
SELECT ticket_price
FROM booking
WHERE passenger_id = 100;
Services
Query_10.sql
lab3-8.sql 38 ms
lab3-15.sql 403 ms
lab3-1.sql 4 s
Query_11.sql
lab3-11.sql 327 ms
Query_2.sql
lab3-5.sql 779 ms
Output
Result 2-2
QUERY PLAN
Index Only Scan using idx_booking_covering on booking (cost=0.14..8.16 rows=1 width=7) (actual time=0.614..0.615 rows=1 loo
Index Cond: (passenger_id = 100)
Heap Fetches: 0
Planning Time: 1.148 ms
5 rows
Databas > queries > lab3-15.sql
Automatic Synchronisation of postgres@localhost
8:22 CRLF UTF-8 4 spaces
Non-commercial use


Databas
master
Database Explorer
postgres@localhost
Family Budget Tracker
public
tables
app_user
columns
id bigint (auto increment)
email text
full_name text
password_hash text
created_at timestamp with tim
keys
indexes
checks
category
columns
lab3-6.sql
lab3-8.sql
lab3-10.sql
lab3-11.sql
lab3-12.sql
lab3-13.sql
lab3-14.sql
lab3-15.sql
Query_2.sq
Tx: Auto
Playground
postgres.public
postgres@localhost
-- 17.
CREATE UNIQUE INDEX idx_airline_code_unique
ON airline(airline_code);
SELECT
airline_code,
COUNT(*) AS duplicate_count
FROM airline
GROUP BY airline_code
HAVING COUNT(*) > 1;
SELECT *
FROM airline
ORDER BY airline_code;
CREATE UNIQUE INDEX idx_airline_code_unique
ON airline(airline_code);
[23505] ОШИБКА: создать уникальный индекс "idx_airline_code_unique" не удалось
Подробности: Ключ (airline_code)=(UNK) дублируется.
Services
Tx
Query_10.sql
lab3-8.sql 38 ms
lab3-15.sql 23 ms
lab3-1.sql 4 s
Query_11.sql
lab3-11.sql 327 ms
Query_2.sql
lab3-5.sql 779 ms
Output
postgres.public.airline
Tx: Auto
DDL
CSV
airline_id
airline_code
airline_name
airline_country
created_at
update_
1 22 2AK6 HGZ PHILIPPINES 2024-03-13 2023-10-2
2 47 AYDN DNU RUSSIA 2024-01-22 2023-09-0
3 43 AYKM KMA FRANCE 2023-11-21 2023-05-3
4 18 AYRA KIQ CHINA 2024-01-13 2023-07-1
54 rows
Databas > queries > lab3-15.sql
13:1 CRLF UTF-8 4 spaces
Non-commercial use